

Markscheme

Mathematics: analysis and approaches
Practice question bank

125. (a)

8 people are to be seated around a round table. Rotations of the same arrangement are considered identical, but reflections (mirror images) are considered different.

Find the number of distinct seating arrangements.

for circular arrangements of n distinct objects where only rotations are equivalent, the count is $(n - 1)!$ *(M1)* **A1**

$$(8 - 1)! = 7! \quad \text{span style="float: right;">*(M1)*$$

$$= 5040 \quad \text{span style="float: right;">**A1**$$

[5 marks]